[PZoom Deep Dive 9.1] Prove that $\psi(a)$ escapes the image of ϕ
Let $x \in S$. We show that $\phi(x) \neq \psi(a)$:

- If $x < a$, the minimality of a ensures that $\phi(x) = \psi(x)$. Since ψ is order-preserving and $x < a$, we get
$$\phi(x) = \psi(x) <' \psi(a).$$
Thus $\phi(x) \neq \psi(a)$.
- If $x = a$ then $\phi(a) \neq \psi(a)$ by the choice of a.
- If $a < x$ then since ϕ is order-preserving,
$$\phi(a) <' \phi(x).$$
Combining this with our assumption $\psi(a) <' \phi(a)$ yields $\psi(a) <' \phi(x)$ by transitivity of the order in S'. Thus again $\phi(x) \neq \psi(a)$.

Therefore no element $x \in S$ satisfies $\phi(x) = \psi(a)$. Hence
$$\psi(a) \notin \mathrm{im}(\phi).$$

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